Satwik Mishra

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Education

Indian Institute of Information Technology, Tiruchirappalli (IIIT Trichy) **Nov 2020 – Apr 2024**
B.Tech, Electronics and Communication Engineering | Institute of National Importance *Tiruchirappalli, India*

Technical Skills

Languages: JavaScript, TypeScript, Java, Python, SQL, C++
Backend: Node.js, Express, Spring Boot, REST APIs, WebSockets, OAuth 2.0/JWT, Microservices
Frontend: React (ReactJS), Redux, Next.js, HTML5, CSS3, Tailwind, Vite
Data: PostgreSQL, Oracle (PL/SQL), MongoDB, Supabase, Firebase
Cloud & DevOps: AWS (EC2, Lambda), Cloudflare (Workers, Pages, D1), CI/CD (GitHub Actions), Git, Postman
Practices: Data Structures & Algorithms, Distributed Systems, Full-stack web application development, Agile methodologies, Software Development Life Cycle (SDLC)

Projects

Career Page Monitor | *Node.js, Puppeteer, PostgreSQL (Supabase), React, Cloudflare Workers, GitHub Actions* **2026**

- Built a scraping system for **120+ company career pages**, with shared adapters for **6 ATS platforms** (Workday, Greenhouse, Lever, Ashby, SmartRecruiters, Personio) and **headless-browser** scrapers for custom sites.
- Orchestrated **hourly scheduled runs** via a **Cloudflare Worker cron** dispatching **GitHub Actions** workflows, replacing GitHub's native scheduler after its triggers proved unreliable.
- Deduplicated postings in **PostgreSQL** on a **composite key** (company, job ID), triggering **Telegram alerts** only for new openings so referrals can be requested within the hour.
- Built a **React** dashboard (Vite, Tailwind) on **Cloudflare Pages** behind **Cloudflare Access**, tracking each job's application status from interested to referral requested to applied.
- Designed a **data-retention lifecycle** (7 days visible, then archived, then purged at ∼30 days) to stay within Supabase's **500 MB** free tier.

Chunky Drive — Distributed File Storage | *React, TypeScript, Cloudflare Workers, D1* **2024 – 2026**

- Built a Google-Drive-style app with **unlimited storage** on a third-party file network: the browser splits files into **8 MB chunks** and verifies each with **SHA-256**, plus a whole-file hash, on reassembly.
- Served the app from a **Cloudflare Worker** that holds storage credentials and **streams chunks without buffering**, moving **3 in parallel** with **retry and backoff**; **D1** stores only the chunk map (∼25 MB per TB).
- Made transfers **pausable and resumable** by persisting verified chunks in **IndexedDB**, surviving closed tabs and dropped connections, with **Google OAuth + email OTP** auth and a **cron** cleaning abandoned uploads.

Experience

Tata Consultancy Services **Nov 2024 – Present**
Systems Engineer — Innovator (TCS's top engineering hiring track)

- Designed and shipped **PDN Tracker**, an internal **deployment-management platform** (React, **Spring Boot**, **Oracle PL/SQL**) replacing a deprecated vendor tool, modeling release lifecycles as a **state machine** with rollback and **audit logging**; serves **500+ daily active users**.
- Implemented **session-based authentication** and **role-based access control (RBAC)** for secure cross-team access, reducing login failures by **35%**.
- Migrated **10+ legacy SOAP services** on **TCS BaNCS** (core banking platform used by banks worldwide) to **REST APIs**, tracing each transaction's **call chain and parameter derivation** across UI, middleware, and **PL/SQL stored procedures** in a **30+ GB monolithic codebase** to define **API contracts** with behavioral parity.
- Re-architected synchronous transaction callbacks into **asynchronous, multi-threaded** processing, cutting response latency from **5s+ to 250 ms**.
- Delivered client **change requests** on the BaNCS workflow, implementing new features and business-rule changes in the **Java** backend.
- Performed **root-cause analysis** on production incidents across application logs, Oracle data, and PL/SQL procedures, resolving the **data-integrity** issues behind the majority of tickets.

Achievements & Activities

LeetCode Knight (achieved in the year 2023, rating **1882**) — 500+ problems solved
Coordinator, Coding Club & Web Dev Club, IIIT Trichy — Organized coding contests, workshops, and hackathons.
Football — Midfielder for the CBSE cluster team, competed at the **state level**.